RITU

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M.Tech Computer Science & Engineering student with a research-driven background in Artificial Intelligence, Machine Learning, and Explainable AI (XAI), complemented by hands-on full-stack development experience across Java, Python, Spring Boot, React, and MySQL. Currently authoring peer-reviewed research on interpretable ensemble learning for healthcare prediction, with work submitted to IEEE Journal of Biomedical and Health Informatics and presented at IEEE-ICAISS 2026. Proven ability to translate academic research into practical, real-world applications — from secure encryption systems to full-stack web platforms — combining strong analytical thinking with a builder's mindset for scalable, reliable software solutions.

EDUCATION

- **SRM Institute of Science & Technology Ghaziabad** **2025-2027**
  Master of Technology [Computer Science & Engineering] [CGPA: 7.78/10]
- **Chhattisgarh Swami Vivekanand Technical University** **2020-2024**
  Bachelors of Technology [Computer Science & Engineering] [CGPA: 8.02/10]
- **Govt Sarvodaya Girls S S S No 1 Narela** **2019-2020**
  Intermediate [Science] [Percentage: 82.5%]

RESEARCH

- **IEEE Journal of Biomedical and Health Informatics** **[Submitted] [Aug 2026]**
  X-Medic: An Explainable Stacking Ensemble Framework for Disease Prediction Using Adaptive Resampling
  - Developed X-Medic, a stacking-based ML ensemble combining LR, SVM, RF, KNN, XGBoost, and LightGBM to address the lack of accuracy, interpretability, and reliability in existing multi-disease prediction models.
  - Achieved up to **97.87% F1-score** across five healthcare datasets and **99.30% accuracy** on an independent 25,000-patient external validation cohort, outperforming individual baseline classifiers.
  - Enhanced clinical trustworthiness via PCA/t-SNE-based dimensionality reduction, XAI-based interpretability, and a confidence-aware prediction mechanism for uncertainty-aware clinical decision support.
- **IEEE-ICAISS 2026** **[Presented] [Apr 2026][Link]**
  Stability Analysis of SHAP and LIME Under Feature Correlation for Migraine Prediction Using Deep Learning
  - Analyzed the stability of SHAP and LIME explainability techniques on a Random Forest migraine prediction model under correlated/redundant clinical features using synthetic feature injection.
  - Quantified explanation stability via Spearman rank correlation, finding LIME more robust to feature correlation (0.93) than SHAP (0.80), highlighting reliability trade-offs between local and global interpretability methods.

PROJECTS

- **ParkNexa – Smart Parking Reservation System** **[Web Link] [GitHub Link]**
  PHP | MySQL | PDO | HTML5 | CSS3 | JavaScript | XAMPP
  - Developed a responsive parking reservation system for location search, slot availability, and booking.
  - Implemented authentication, registration, reservations, confirmations, cancellations, and booking history.
  - Built an admin dashboard to manage parking locations, slots, users, and reservations using CRUD operations.
  - Integrated MySQL with PHP/PDO for secure data storage, retrieval, and transaction-based reservation handling.
  - Implemented client-side validation, search/filter functionality, responsive UI, and real-time slot availability.
- **RailGenie | Railway Maintenance Optimization** **[GitHub Link]**
  React | Python | FastAPI | Optimization
  - Developed a railway maintenance scheduling system using React and FastAPI.
  - Implemented priority-based scheduling, safety buffers, and conflict detection for train movements.
  - Added what-if simulation, diagnostics, scoring, and analytics for optimized maintenance planning.
  - Developed conflict detection and schedule scoring to improve maintenance planning and operational efficiency.

SKILLS

- **Programming & Embedded Systems:** Java, Python, PHP, HTML5, CSS3, JavaScript
- **Languages and Frameworks:** React, Spring Boot, FastAPI, Node.js
- **Databases & Tools:** MySQL, PDO, XAMPP, Git, VS Code
- **Optimization:** Scheduling Optimization, Constraint-Based Optimization
- **Generative AI & LLM Tools:** ChatGPT, Claude AI, Google Gemini, GitHub Copilot, Perplexity AI
- **Soft Skills:** Self motivated, Self driven & Solution oriented mindset, Communication & Coordination
- **Language:** Hindi[Native Proficiency], English [Professional Working Proficiency]

CERTIFICATION

- **Full Stack Developer** **[TAP Academy, Bangalore][Sep 2024][Link]**
  - Learned Core Java and Advanced Java for backend development and full-stack web development.
  - Gained hands-on experience with HTML, CSS, JavaScript, SQL, and Node.js.